

Nitin Avula

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Education

University of Florida

Gainesville, FL

M.S. in Computer & Information Science & Engineering (GPA: 3.7/4.0)

Aug 2024 – May 2026

Coursework: Data Structures & Algorithms, Distributed Systems, Database Systems, Software Engineering, Machine Learning

Technical Skills

Programming: Python, Java, C++, Go, C#, TypeScript, SQL, Rust

Frameworks & Libraries: React, Spring Boot, Flask, FastAPI, Node.js, Express.js, Pandas, NumPy

Backend & Systems: Microservices, RESTful Services, Event-Driven Architecture, Concurrency, Distributed Computing, API Security, Algorithm Development

Cloud & DevOps: AWS (EC2, CloudWatch), Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions

Data & Storage: PostgreSQL, MySQL, MongoDB, NoSQL Databases, Vector Search, Caching Layers, Index Optimization

Specialized Areas: Machine Learning, Data Engineering, Information Systems, Embedded Systems Basics

Engineering Practices: Agile Development, Debugging, Testing, Code Reviews, Technical Documentation, Monitoring, Observability

Core Strengths: Scalable System Design, Performance Optimization, Distributed Processing, Fault Tolerance

Experience

Software Engineer Intern

Oct 2023 – Jun 2024

Iconize Technologies

- Implemented backend microservices for a genomic analytics platform, constructing modular RESTful APIs that improved researcher query turnaround time by **30%**.
- Optimized PostgreSQL performance through index tuning, SQL execution plan analysis, and connection pooling, reducing average response latency by **40%**.
- Architected resilient workflow orchestration services with validation, retry handling, and fault isolation, lowering pipeline failure rates by **25%**.
- Authored internal **technical documentation** for API workflows and system architecture, accelerating onboarding and cross-team knowledge sharing.
- Performed root-cause analysis and production **debugging**, improving service reliability and reducing recurring incidents.
- Automated containerized deployments using Jenkins pipelines, rolling updates, readiness checks, and health monitoring, cutting deployment time from 20 to **6 minutes**.

Data Science Intern

May 2023 – Jul 2023

- Delivered a production-grade traffic prediction system using ensemble learning (Random Forest, Gradient Boosting), improving forecast accuracy to **92%**.
- Deployed a low-latency inference service supporting concurrent API requests with optimized memory allocation and batching, sustaining **P95 response times below 200ms**.
- Developed scalable data preprocessing pipelines using **SQL** and Python to clean, transform, and validate high-volume datasets.
- Established real-time observability with AWS CloudWatch metrics, logs, and alarms, reducing mean time to detect anomalies by **35%**.
- Automated model validation and drift detection workflows, maintaining prediction stability under evolving data patterns.

Projects

Microservice Transaction System — Spring Boot & Kafka

- Architected an asynchronous transaction processor using event streaming and producer-consumer patterns, sustaining **50 events/sec** under load testing.
- Ensured durability and consistency through idempotent consumers, transactional writes, and persistent storage integration.
- Applied efficient **algorithm development** techniques to handle message ordering and deduplication guarantees.
- Implemented integration tests simulating failure recovery, replay scenarios, and out-of-order event handling.

Scalable AI Fitness Platform (Full Stack System)

- Architected a session-aware backend supporting 200+ concurrent users, reducing redundant AI calls by **35%** using caching and deduplication.
- Engineered resilient REST endpoints with rate limiting, schema validation, and structured logging to improve service reliability.
- Developed a real-time React interface with progress tracking, synchronized state management, and responsive feedback loops.